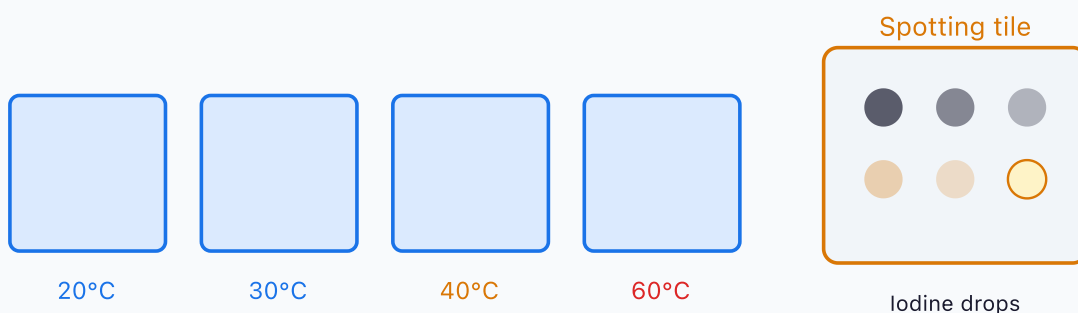


Edexcel B A-Level Biology — Core Practicals

Table of Contents

1. CP1: A Factor Affecting the Initial Rate of an Enzyme-Controlled Reaction (Temperature, Amylase)
2. CP2: Light Microscopy with Stage and Eyepiece Micrometers; Drawing Cells from a Specialised Tissue
3. CP3: Root Tip Squash to Show the Stages of Mitosis
4. CP4: Effect of Sucrose Concentration on Pollen Tube Growth
5. CP5: Effect of Temperature on Beetroot Membrane Permeability
6. CP6: Determining the Water Potential of Plant Cells
7. CP7: Dissection of an Insect to Show the Gas Exchange System
8. CP8: Factors Affecting Water Uptake by Plant Shoots (Potometer)
9. CP9: Factors Affecting the Rate of Aerobic or Anaerobic Respiration (Respirometer)
10. CP10: Effect of Different Wavelengths of Light on the Rate of Photosynthesis
11. CP11: Separating Chloroplast Pigments by Chromatography
12. CP12: Rate of Growth of Bacteria in Liquid Culture
13. CP13: Isolating Bacteria from a Mixed Culture by Streak Plating
14. CP14: Effect of Gibberellin on Amylase Production in Germinating Cereals (Starch Agar Assay)
15. CP15: Effect of Different Sampling Methods on Estimates of Population Size
16. CP16: Effect of One Abiotic Factor on the Distribution or Morphology of One Species

CP1: A Factor Affecting the Initial Rate of an Enzyme-Controlled Reaction (Temperature, Amylase)



Amylase + starch → sample into iodine every 30 s
End-point: iodine stays yellow-brown (no starch remaining)

$$\text{Rate} = 1 / \text{time to end-point}$$

Apparatus

- Amylase solution, starch solution (1%)
- Iodine solution, white spotting tile
- Water baths at 20, 30, 40, 50, 60 °C
- Test tubes, pipettes, stopwatch, thermometer

Method

1. Place amylase and starch in separate test tubes in a water bath. Equilibrate for 5 minutes.
2. Mix amylase and starch. Start the stopwatch immediately.
3. Every 30 seconds, remove a drop and place on a spotting tile with iodine solution.
4. Record the time at which iodine no longer turns blue-black (all starch digested).
5. Repeat at each temperature (3 repeats). Calculate rate = 1/time.

Safety

- Iodine is an irritant and stains skin — wear gloves.
- Care with hot water baths — use test tube holders.

Analysis

- Plot rate ($1/\text{time}$) against temperature. Expect a bell-shaped curve.
- Optimum temperature for amylase is around 37°C . Above this, the enzyme denatures.
- Below the optimum, rate increases as kinetic energy increases (more successful collisions between enzyme and substrate).
- Other factors (pH, enzyme concentration, substrate concentration, inhibitor concentration) can be investigated in the same way, changing only that one factor.

Sources of Error

Error	Effect	Improvement
Tubes not equilibrated	Reaction starts at wrong temperature	Equilibrate both tubes for 5 minutes
Sampling interval too long	Miss the exact end-point	Sample every 15–30 seconds near end-point
Judging colour change subjectively	Inconsistent end-point	Use same person for all judgements; use colorimeter
Contamination of iodine wells	False readings	Use fresh iodine drop for each sample

Variables

Variable Type	Variable	How Controlled
Independent	Temperature	Set by water baths (use thermometer to verify)
Dependent	Time for starch digestion (then rate = $1/t$)	Timed with stopwatch; iodine test for end-point
Controlled	Volume and concentration of amylase and starch	Use same pipette and stock solutions

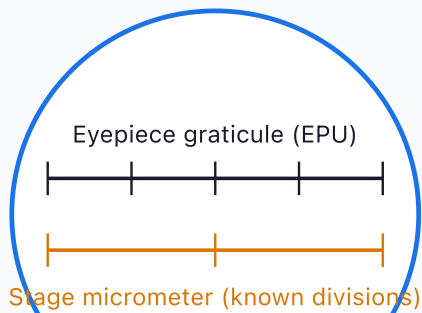
Controlled	pH	Use buffer solution (pH 7)
Controlled	Equilibration time	5 minutes for both solutions before mixing

Expected Results

Rate increases with temperature up to the optimum (~37 °C for salivary amylase) because higher kinetic energy means more frequent and energetic collisions between enzyme and substrate, increasing the likelihood of forming enzyme-substrate complexes. Above the optimum, hydrogen and ionic bonds holding the tertiary structure break, the active site changes shape (denaturation), and the substrate can no longer bind. At very high temperatures (60+ °C), the enzyme is completely denatured and the rate drops to zero. The Q_{10} value (the factor by which rate increases per 10 °C rise) is typically around 2 for enzyme reactions below the optimum.

Exam Tip: Calculate rate as $1/t$ (where t is the time for starch to be completely digested). Strictly, this is a mean rate over the whole reaction, used as an estimate of the initial rate; a truer initial rate comes from following the reaction continuously (e.g. with a colorimeter) and taking the gradient of a tangent at $t = 0$. Explain the decrease in rate above the optimum in terms of *denaturation* — the active site changes shape permanently and can no longer form enzyme-substrate complexes. Never say the enzyme is "killed" (it is a protein, not alive) or that it is "destroyed" (the molecule still exists, but its shape has changed irreversibly).

CP2: Light Microscopy with Stage and Eyepiece Micrometers; Drawing Cells from a Specialised Tissue



Calibration

$1 \text{ EPU} = \text{stage distance} \div \text{number of EPUs}$

$\text{Actual size} = \text{EPUs} \times \text{value of 1 EPU}$

$\text{Magnification} = \text{image size} \div \text{actual size}$

Recalibrate for every objective lens

Apparatus

- Light microscope with $\times 4$, $\times 10$ and $\times 40$ objectives
- Eyepiece graticule, stage micrometer slide (e.g. 1 mm divided into 100 divisions of $10 \mu\text{m}$)
- Prepared slide of a specialised tissue (e.g. palisade mesophyll in a leaf section, ciliated epithelium, or a blood smear)
- Sharp HB pencil, plain paper, ruler

Method

1. Fit the eyepiece graticule into the eyepiece and put the stage micrometer on the stage.
2. Focus on the stage micrometer using the lowest-power objective, then line up its scale with the graticule scale.
3. Count how many eyepiece units (EPUs) match a known distance on the stage micrometer. Calculate the length of one EPU. Repeat for each objective lens.
4. Replace the micrometer with the specimen slide. Focus using the coarse then fine focus, starting on low power.
5. Measure several cells in EPUs and convert to μm using the calibration for that objective.
6. Draw a **small number of cells** (e.g. 3–5) from the tissue at high power, showing their true shape and relative proportions.

Safety

- Handle glass slides and coverslips carefully; report breakages.

- Never use direct sunlight as the light source (eye damage).

Analysis

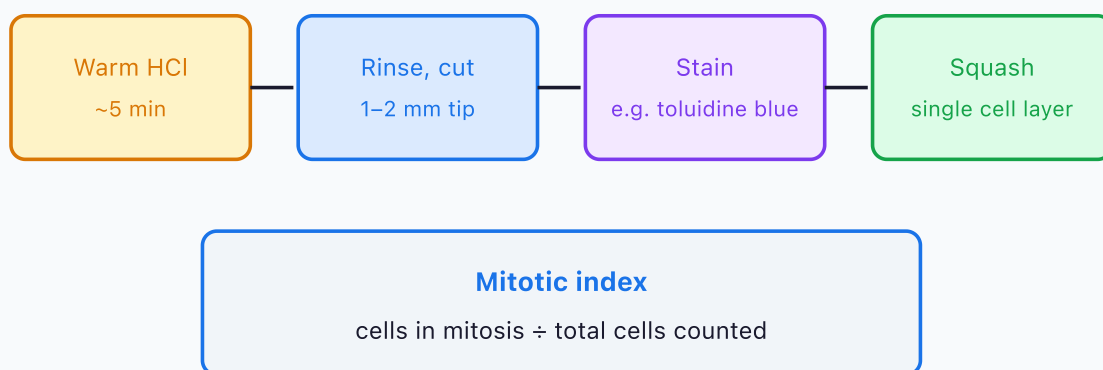
- Actual size = number of EPUs $\times$ length of one EPU at that magnification.
- Magnification = size of image $\div$ actual size of object (keep units the same; 1 mm = 1000 μ m).
- Calculate a mean cell size from several measurements.
- Relate the features seen to the tissue's function (e.g. many chloroplasts in palisade cells).

Sources of Error

Error	Effect	Improvement
Using the calibration from a different objective	Measurements wrong by a factor of the magnification change	Calibrate separately for each objective lens
Measuring a single cell	Not representative	Measure several cells and calculate a mean
Cells cut at different angles in a section	Apparent size varies	Measure only cells cut in the same plane

Exam Tip: Scientific drawings: use a sharp pencil, clear continuous lines with no sketching or shading, draw only what you see, keep proportions accurate, label with ruled lines that do not cross, and give a title and magnification or scale bar. The eyepiece graticule has no units until it is calibrated against the stage micrometer.

CP3: Root Tip Squash to Show the Stages of Mitosis



Apparatus

- Garlic or onion root tips (grown over water), 1 mol dm⁻³ HCl in a water bath at about 55–60 °C
- Toluidine blue or acetic orcein stain, distilled water, watch glass
- Slides, coverslips, scalpel, forceps, mounted needle, filter paper, light microscope

Method

1. Cut the end 5 mm from a few root tips and place in warm HCl for about 5 minutes.
2. Rinse in cold water and blot dry. Put one tip on a slide and keep only the terminal 1–2 mm (meristem).
3. Add a drop of stain, break the tissue up with a mounted needle and leave for about 2 minutes.
4. Add a coverslip, cover with filter paper and press down firmly and vertically to spread the cells into a single layer.
5. Find the meristem under low power, then use high power to identify cells in interphase, prophase, metaphase, anaphase and telophase.
6. Count the cells in each stage in several fields of view.

Safety

- HCl is an irritant — wear eye protection.
- Stains mark skin — wear gloves. Take care with the scalpel and when pressing the coverslip.

Analysis

- Mitotic index = number of cells in mitosis $\div$ total number of cells.
- The proportion of cells in each stage reflects the relative time spent in that stage.
- Draw a few cells at different stages, showing chromosomes as clear lines.

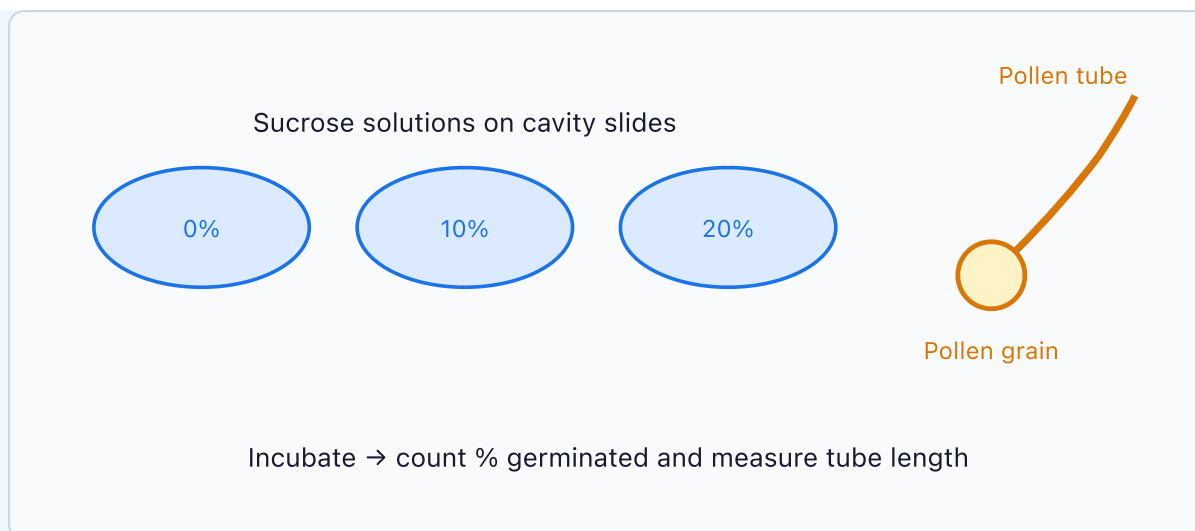
Sources of Error

Error	Effect	Improvement
Squash too thick	Cells overlap; chromosomes unclear	Use only the tip; macerate well; squash firmly without twisting
Tissue not from the meristem	Few dividing cells	Cut the terminal 1–2 mm only
Stages misidentified	Inaccurate counts	Agree criteria for each stage; count many cells

Exam Tip: Know why each step is done: HCl breaks down the middle lamella so the cells separate (and stops division); the stain binds to DNA to show chromosomes; squashing gives one layer of cells so light passes through. Be able to identify stages from photographs and to calculate a mitotic index.



CP4: Effect of Sucrose Concentration on Pollen Tube Growth



Apparatus

- Flowers with ripe anthers (e.g. *Impatiens*, lily, *Tradescantia*)
- Sucrose solutions of different concentrations (e.g. 0, 5, 10, 15, 20%)
- Cavity slides, coverslips, paintbrush, Petri dish lined with damp filter paper (humid chamber)
- Light microscope with calibrated eyepiece graticule, stopwatch, incubator or warm place

Method

1. Put a drop of each sucrose solution into a separate labelled cavity slide.
2. Gently tap or brush pollen from ripe anthers onto the surface of each drop, spreading it thinly.
3. Add a coverslip and place the slides in a humid chamber at a constant temperature (e.g. 25 °C) so the drops do not dry out.
4. After a fixed time (e.g. 30–60 minutes), count the number of grains that have produced a tube in a field of view, and measure tube lengths with the eyepiece graticule.
5. Count at least 50 grains per slide and repeat for each concentration.

Safety

- Pollen can trigger allergies and hay fever — avoid inhaling it and wash hands afterwards.
- Handle slides and coverslips carefully.

Analysis

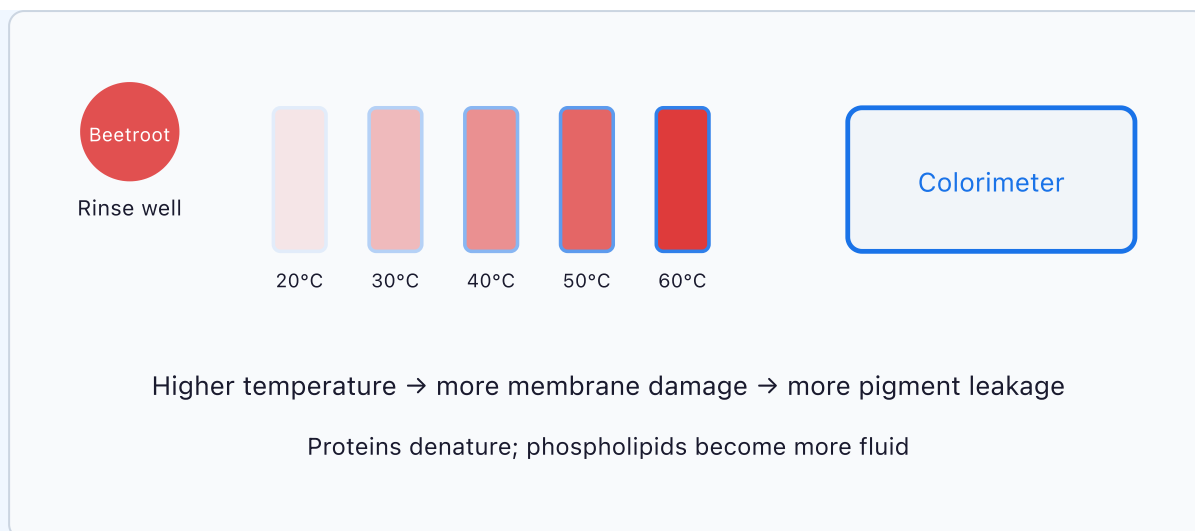
- Percentage germination = $\frac{\text{grains with a tube}}{\text{total grains counted}} \times 100$. A grain is usually counted as germinated if its tube is longer than the grain's diameter.
- Plot percentage germination and mean tube length against sucrose concentration. Growth is usually greatest at an intermediate concentration.
- Explain: sucrose provides a respiratory substrate for tube growth, and affects water potential — too dilute and grains may take up water and burst; too concentrated and they lose water by osmosis.

Sources of Error

Error	Effect	Improvement
Drops dry out	Concentration rises during incubation	Use a humid chamber; keep coverslips on
Pollen clumps	Grains hard to count; growth affected	Spread pollen thinly with a fine brush
Pollen from flowers of different ages	Different viability	Use freshly opened flowers of the same age and species

Exam Tip: Link the practical to fertilisation in flowering plants: the pollen tube grows down the style by secreting digestive enzymes, carrying the male nuclei to the embryo sac for **double fertilisation**. Explain the effect of sucrose using both respiration (energy for growth) and water potential (osmosis).

CP5: Effect of Temperature on Beetroot Membrane Permeability



Apparatus

- Fresh beetroot, cork borer, scalpel, white tile
- Test tubes, distilled water, water baths (20–70 °C)
- Colorimeter and cuvettes, thermometer

Method

1. Cut uniform beetroot discs. Rinse thoroughly to remove surface pigment.
2. Place one disc in each tube with 5 cm³ distilled water. Incubate at each temperature for 20 min.
3. Remove beetroot. Measure absorbance using colorimeter.
4. Repeat 3 times per temperature. Calculate means.

Safety

- Sharp tools — cut on white tile away from body.
- Hot water — use tongs for tubes.

Analysis

- Plot mean absorbance against temperature.
- Absorbance increases sharply above ~40 °C as membrane proteins denature and phospholipid bilayer loses integrity.

Sources of Error

Error	Effect	Improvement
Unequal disc sizes	Different surface area	Use same cork borer and trim uniformly
Insufficient rinsing	Surface pigment inflates readings	Rinse until water is clear
Fingerprints on cuvette	Inaccurate absorbance	Handle cuvette by top; wipe with tissue

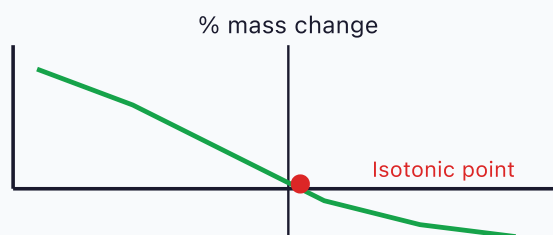
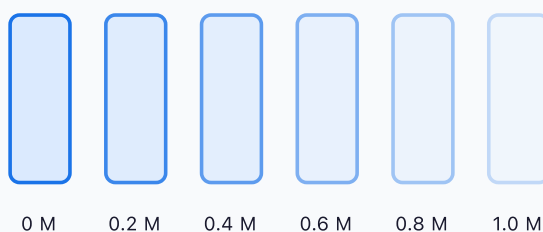
Exam Tip: Explain results using the fluid mosaic model: membrane proteins denature (tertiary structure disrupted by heat), creating gaps. Phospholipids gain kinetic energy and the bilayer becomes more permeable, allowing betacyanin to diffuse out.

CP6: Determining the Water Potential of Plant Cells



Potato cylinders
(weigh before)

Sucrose concentrations



Gains mass
(water enters by osmosis)

Loses mass
(water leaves)

Apparatus

- Potato, cork borer, scalpel, ruler
- Sucrose solutions at 0, 0.2, 0.4, 0.6, 0.8, 1.0 mol dm⁻³
- Test tubes or boiling tubes, balance (to 0.01 g)
- Paper towels, labels, stopwatch

Method

1. Cut potato cylinders of equal length using a cork borer. Blot dry and weigh each one.
2. Place one cylinder in each sucrose concentration. Leave for 30–60 minutes.
3. Remove, blot dry gently, reweigh.
4. Calculate % change in mass: (final – initial) / initial × 100.
5. Repeat 3 times per concentration.

Safety

- Care with sharp cutting tools — cut on a white tile.
- Sucrose solutions are non-hazardous.

Analysis

- Plot % change in mass (y-axis) against sucrose concentration (x-axis).
- Where the line crosses zero = isotonic point (water potential of potato cells equals external solution).
- Read off the sucrose concentration at 0% change in mass, then convert it to a water potential (in kPa) using a conversion table or graph. This value is the water potential of the tissue.
- Below this concentration: cells gain water (hypotonic solution). Above: cells lose water (hypertonic solution).

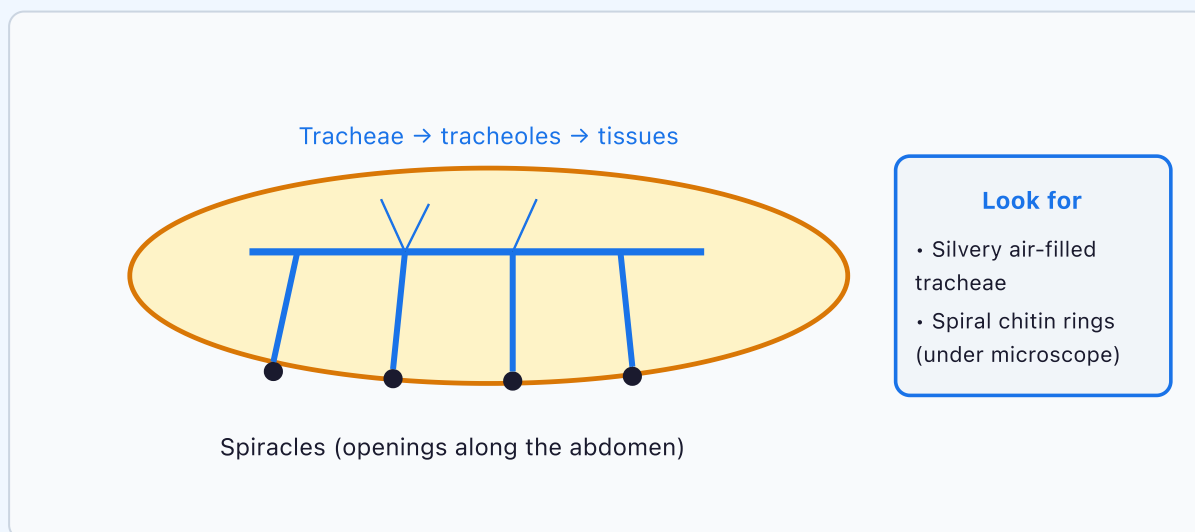
Sources of Error

Error	Effect	Improvement
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Cylinders different sizes	Different surface areas affect rate	Use same cork borer; trim to equal length
Inconsistent blotting	Variable surface water affects mass	Blot once each side; same technique
Not enough time left in solution	Equilibrium not reached	Leave for at least 30 minutes
Potato from different parts of tuber	Different water potential	Use cylinders from same area of same potato

Exam Tip: The isotonic point (where % mass change = 0) gives the water potential of the potato cells. Water moves from high water potential to low water potential by osmosis. Always use *percentage* change in mass, not absolute change, so different starting masses are accounted for.

CP7: Dissection of an Insect to Show the Gas Exchange System



Apparatus

- Freshly killed large insect (e.g. locust or cockroach) from a reputable supplier
- Dissecting board or wax-bottomed dish, dissecting pins, fine scissors, scalpel, forceps, mounted needle
- Water or saline, hand lens or stereo (dissecting) microscope, slides, coverslips, light microscope

Method

1. Examine the outside of the insect with a hand lens and locate the **spiracles** along the sides of the thorax and abdomen.
2. Pin the insect dorsal side up. Remove the wings and legs if needed.
3. Using fine scissors, cut carefully along the dorsal midline of the abdomen from the tip towards the head, keeping the scissor points up so internal organs are not damaged.
4. Pin back the body wall on each side and cover the insect with water or saline so the tracheae float and show up as **silvery tubes** (they contain air).
5. Trace the tracheae from the spiracles inwards, noting how they branch into smaller tubes.
6. Remove a small piece of trachea, mount it in a drop of water and view it under the light microscope to see the **spiral rings of chitin** and the fine tracheoles.

Safety and Ethics

- Sharp instruments — cut away from yourself and your fingers; count pins in and out.
- Locust dust and body parts can cause allergies and asthma — anyone with such an allergy should not take part; wear gloves and wash hands.
- Use insects killed humanely by a supplier, use as few as possible, and treat specimens with respect. Dispose of remains as your teacher directs.

Analysis

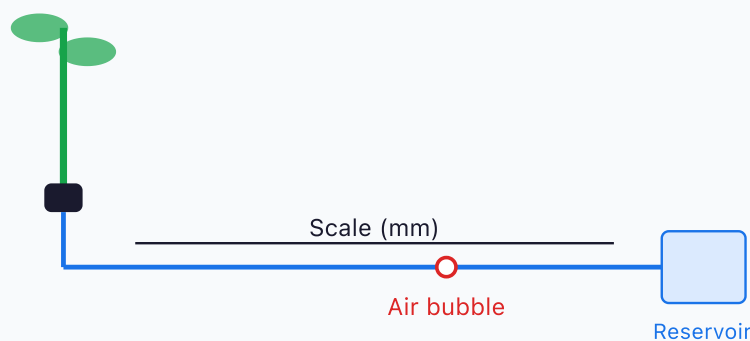
- Draw and label the arrangement of spiracles, tracheae and tracheoles.
- Relate structure to function: spiracles can close to reduce water loss; chitin rings keep tracheae open; tracheoles end close to (or in) respiring cells, giving a short diffusion distance; ventilation movements of the abdomen help move air.

Sources of Error

Error	Effect	Improvement
Cutting too deeply	Tracheae and organs damaged	Keep scissor points angled upwards; make shallow cuts
Specimen dries out	Tracheae collapse and are hard to see	Keep the dissection covered with water or saline
Tracheae hard to distinguish from other tissue	Incorrect identification	Use a stereo microscope; look for air-filled silvery tubes

Exam Tip: Be able to describe the path of oxygen: **spiracle → trachea → tracheole → respiring cell**, by diffusion down a concentration gradient (plus ventilation in larger insects). Ethics questions often ask how to minimise harm: use dead specimens from a supplier, use the minimum number, and consider alternatives such as videos or models.

CP8: Factors Affecting Water Uptake by Plant Shoots (Potometer)



Apparatus

- Potometer, leafy shoot, rubber tubing
- Capillary tube with scale, water reservoir
- Fan, lamp, plastic bag, Vaseline
- Stopwatch, ruler

Method

1. Cut shoot underwater. Assemble potometer underwater to prevent air locks.
2. Seal all joints with Vaseline. Allow 5 min to equilibrate.
3. Introduce an air bubble. Record distance moved per minute.
4. Test under different conditions: windy (fan on), humid (plastic bag over leaves), bright light, darkness.
5. Reset bubble between tests using reservoir. Repeat 3 times per condition.

Safety

- Sharp tools for cutting stems. Keep electrical equipment away from water.

Analysis

- Volume of water taken up = $\pi r^2 \times \text{distance moved by the bubble}$ (r = radius of the capillary bore); rate = volume $\div$ time.

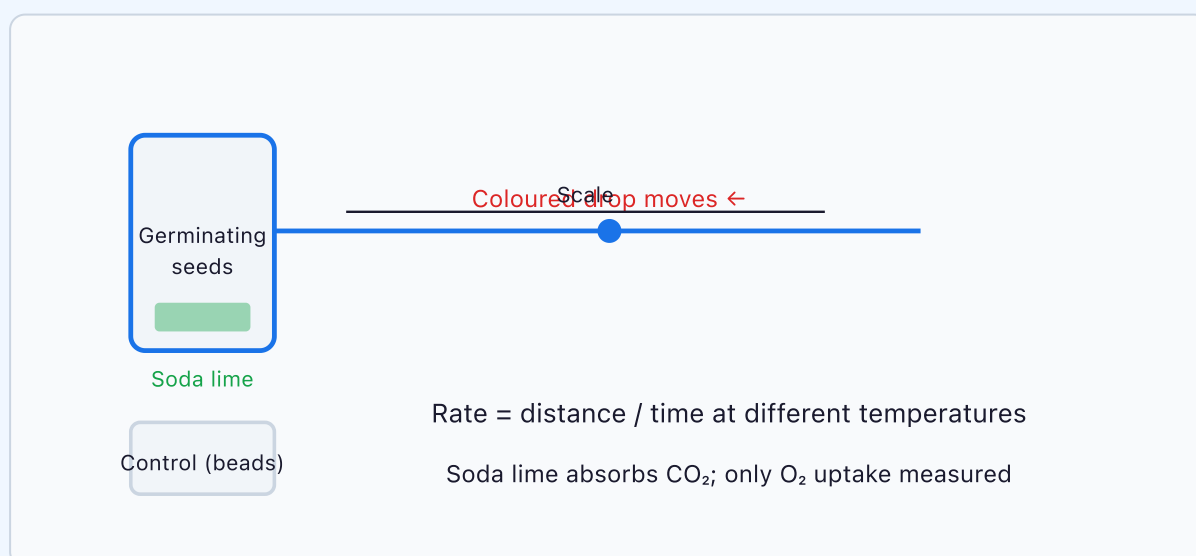
- Rate of water uptake $\approx$ rate of transpiration.
- Higher wind speed, lower humidity, higher temperature, and brighter light all increase transpiration rate.

Sources of Error

Error	Effect	Improvement
Air lock in xylem	No water uptake; bubble doesn't move	Cut stem underwater
Leaks at joints	Water loss not through leaves	Vaseline all connections
Wilting	Stomata close, reducing transpiration	Keep in water; do not delay setup

Exam Tip: A potometer measures water *uptake*, not transpiration directly. Transpiration creates a transpiration pull (cohesion-tension theory), drawing water up the xylem. State how each environmental factor affects stomatal opening and the water potential gradient.

CP9: Factors Affecting the Rate of Aerobic or Anaerobic Respiration (Respirometer)



Apparatus

- Simple respirometer, germinating seeds, glass beads (control)
- Soda lime, cotton wool, capillary tube with scale
- Coloured fluid, water bath, stopwatch

Method

1. Set up respirometer with seeds and soda lime. Control uses glass beads.
2. Seal, place in water bath, equilibrate 5 min.
3. Introduce coloured fluid drop. Record distance moved per minute for 10 min.
4. Repeat at different temperatures.
5. Anaerobic respiration can be investigated with yeast in glucose solution (no soda lime), collecting the CO₂ released in a gas syringe.

Safety

- Soda lime is corrosive — gloves and eye protection.
- If small invertebrates (e.g. woodlice or maggots) are used, keep them on gauze away from the soda lime, keep the experiment short and return them afterwards.

Analysis

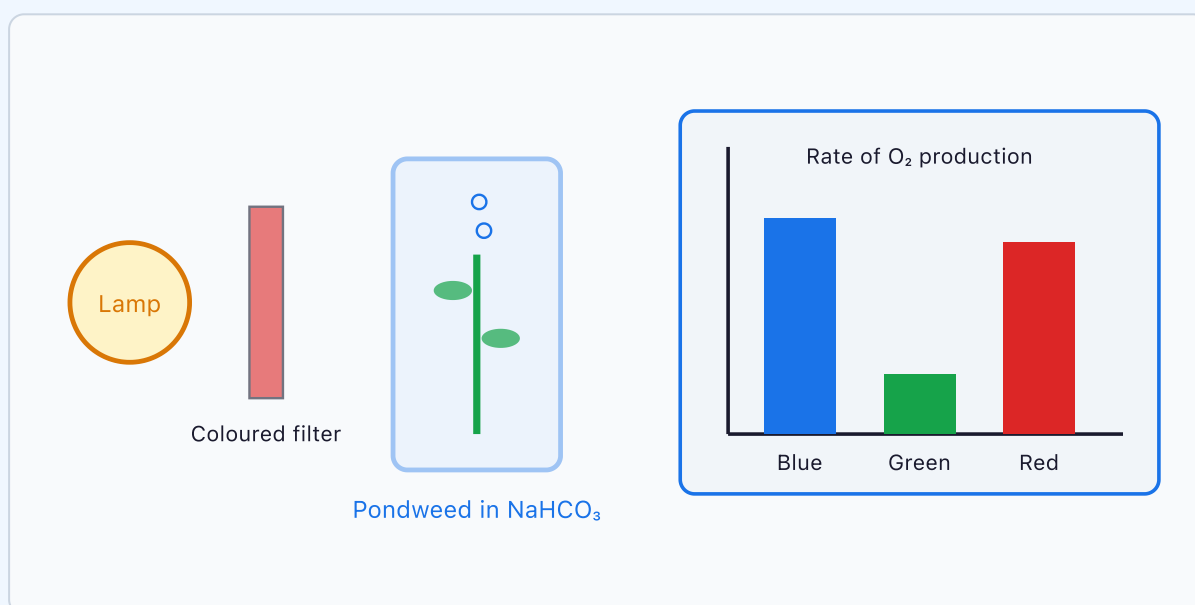
- Rate = distance / time. Plot rate vs temperature.
- Q₁₀ effect: rate roughly doubles per 10 °C (up to optimum).

Sources of Error

Error	Effect	Improvement
Leaks	Fluid doesn't move	Seal with Vaseline
No equilibration	Temperature not correct at start	Wait 5 min before recording
No control	Cannot account for pressure changes	Always include glass bead control

Exam Tip: The control tube with glass beads accounts for changes in atmospheric pressure and temperature that are not due to respiration. Soda lime absorbs CO_2 , so only O_2 consumption causes volume change.

CP10: Effect of Different Wavelengths of Light on the Rate of Photosynthesis



Apparatus

- Pondweed (e.g. *Elodea* or *Cabomba*) in dilute sodium hydrogencarbonate solution, boiling tube or photosynthometer (capillary tube and syringe) to collect gas
- Bench lamp, coloured filters (e.g. red, blue, green) or coloured LEDs, light meter
- Water bath or beaker of water as a heat shield, thermometer, stopwatch, ruler
- Alternative: leaf discs in sodium hydrogencarbonate solution (floating leaf disc method)

Method

1. Place a freshly cut piece of pondweed, cut end up, in a tube of NaHCO_3 solution and let it acclimatise in white light for a few minutes.

2. Place a coloured filter between the lamp and the tube. Keep the lamp at a fixed distance behind a heat shield.
3. Use a light meter to check the light intensity reaching the plant through each filter; adjust the distance so each colour gives the same intensity, if possible.
4. Allow a few minutes to adjust, then collect the gas produced over a fixed time (e.g. 5 minutes) in a photosynthometer, and measure its volume (or count bubbles).
5. Repeat for each colour of light, three times each. Include a dark control (tube covered).

Safety

- Keep the lamp and electrical cables away from water.
- Lamps get hot — do not touch the bulb; use a heat shield.

Analysis

- Calculate the mean rate of gas production for each wavelength (colour).
- Expected: highest rates in **blue** and **red** light and lowest in **green** light, because chlorophylls absorb mainly blue and red light and reflect or transmit green light.
- Compare the results with the absorption spectra of chlorophyll a, chlorophyll b and the carotenoids to explain the **action spectrum**. Some photosynthesis still occurs in green light, because accessory pigments absorb some of it.

Sources of Error

Error	Effect	Improvement
Filters transmit different light intensities	Intensity, not wavelength, may cause the difference	Measure intensity with a light meter and adjust the lamp distance to equalise it
Counting bubbles	Bubbles vary in size	Collect gas and measure its volume in a photosynthometer
Heat from the lamp	Temperature changes the rate	Use an LED lamp or a heat shield; monitor temperature

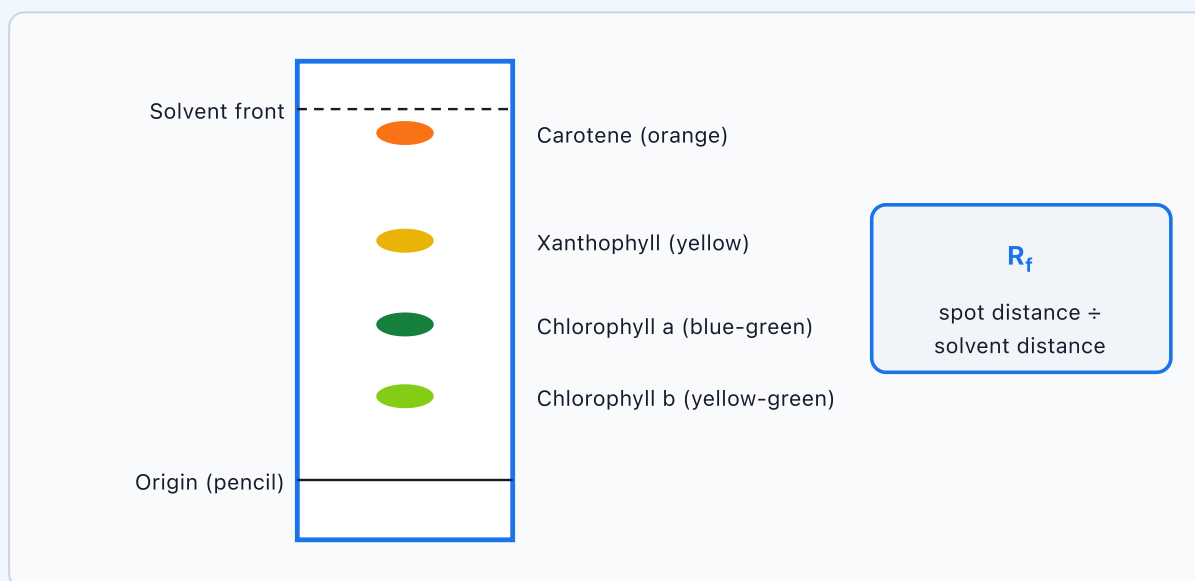
Background light in the room

Plant also receives white light

Work in a darkened room

Exam Tip: Distinguish the **absorption spectrum** (how much light of each wavelength a pigment absorbs) from the **action spectrum** (the rate of photosynthesis at each wavelength). They are similar in shape, which is evidence that the pigments absorb the light used in photosynthesis. Having several pigments lets a plant use a wider range of wavelengths.

CP11: Separating Chloroplast Pigments by Chromatography



Apparatus

- Fresh leaves (e.g. spinach, nettle), scissors, pestle and mortar, sand, propanone (acetone)
- Chromatography paper or a thin-layer chromatography (TLC) plate, fine capillary tube, pencil, ruler
- Chromatography solvent (e.g. a petroleum ether and propanone mixture), boiling tube or tank with a bung or lid

Method

1. Grind a few leaves with a little sand and a small volume of propanone to extract a dark green pigment solution.
2. Draw a pencil origin line about 1.5 cm from the bottom of the paper or plate.
3. Use a capillary tube to put a tiny spot of extract on the origin. Let it dry, then add more on the same spot. Repeat many times to build up a small, concentrated spot.
4. Stand the paper in the solvent, with the **origin above the solvent level**, and close the container.
5. When the solvent front is near the top, remove the paper and immediately mark the solvent front in pencil.
6. Mark the centre of each coloured spot and measure its distance from the origin.

Safety

- Propanone and chromatography solvents are **highly flammable** and harmful to breathe — no naked flames, use a well-ventilated room or fume cupboard, and keep containers closed.
- Wear eye protection and gloves.

Analysis

- $R_f = \text{distance moved by the pigment spot} \div \text{distance moved by the solvent front}$.
- Identify pigments by colour and R_f : carotenes move furthest, then xanthophylls, then chlorophyll a, with chlorophyll b moving least in a typical solvent.
- Compare extracts from different plants or from sun and shade leaves, and explain why having several pigments allows a wider range of wavelengths to be absorbed.

Sources of Error

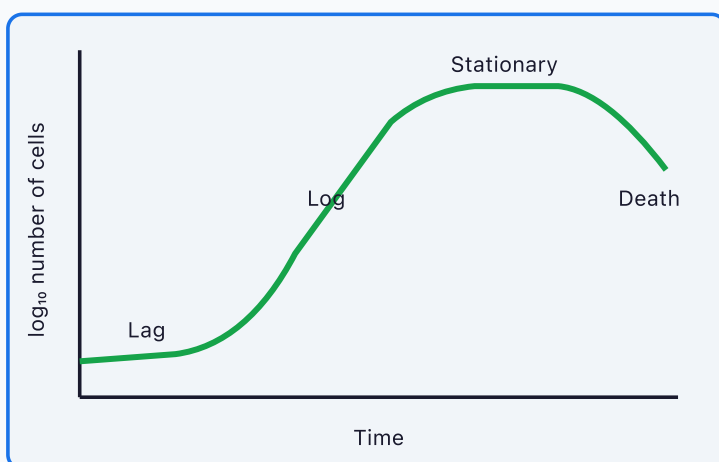
Error	Effect	Improvement
Spot too large or too dilute	Spots smear or are too faint to see	Build up a small spot by repeated drying and re-spotting

Origin below the solvent level	Pigments dissolve into the solvent instead of moving up	Check the solvent depth before inserting the paper
Solvent front not marked immediately	Solvent evaporates; R_f values inaccurate	Mark the front as soon as the paper is removed
Container left open	Solvent evaporates from paper; uneven running	Keep the container sealed during the run

Exam Tip: Always draw the origin line in **pencil** (ink would dissolve and run). R_f values are always less than 1 and depend on the solvent, so compare them only when the same solvent and conditions are used. Pigments with a higher R_f are more soluble in the solvent and/or less strongly attracted to the paper.



CP12: Rate of Growth of Bacteria in Liquid Culture



Measure growth by

- Turbidity (colorimeter)
- Dilution plating (colony counts)
- Cell counts

Apparatus

- Culture of a non-pathogenic bacterium (e.g. *Bacillus subtilis* or *E. coli* K-12), sterile nutrient broth in a flask

- Sterile pipettes, Bunsen burner, disinfectant, incubator or water bath (e.g. 25 °C)
- Colorimeter and cuvettes (for turbidity), and/or sterile agar plates and sterile water for serial dilutions

Method

1. Using aseptic technique, add a small, fixed volume of bacterial culture to a flask of sterile broth. Mix and keep at a constant temperature.
2. Immediately, and then at regular intervals (e.g. every 30–60 minutes), remove a sample aseptically.
3. **Turbidity:** measure the absorbance of each sample in a colorimeter, zeroed with sterile broth. More cells make the broth cloudier, so absorbance rises.
4. **Dilution plating:** make a serial dilution of the sample (e.g. 1 in 10 steps), spread a fixed volume of suitable dilutions on agar plates, incubate, then count the colonies on plates with 30–300 colonies.
5. Continue sampling for long enough to see the curve level off.

Safety

- Use aseptic technique throughout; incubate at no more than 25–30 °C as advised; do not open plates after incubation.
- Disinfect the bench and wash hands; autoclave cultures, plates and pipettes before disposal.

Analysis

- Number of cells in the original sample = number of colonies × dilution factor (per volume plated).
- Plot $\log_{10}$ number of cells (or absorbance) against time and identify the lag, log (exponential), stationary and death phases.
- Mean growth rate constant: $k = (\log_{10}N_t - \log_{10}N_0) \div (0.301 \times t)$, giving the number of generations per unit time. ($0.301 = \log_{10}2$.)

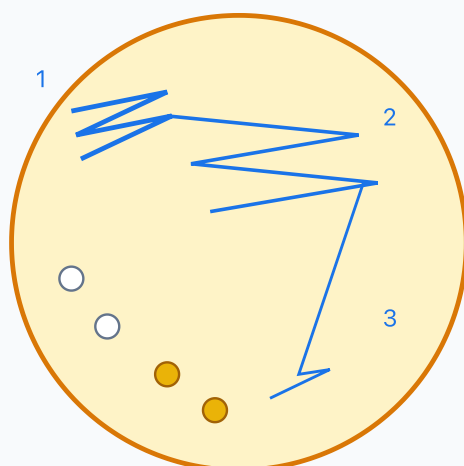
Sources of Error

Error	Effect	Improvement
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Turbidity includes dead cells	Overestimates the number of living cells	Use dilution plating for viable counts
Contamination	Other organisms add to cell numbers	Strict aseptic technique; sterile broth and pipettes
Colonies merge on plates	Undercounts	Use a higher dilution; count only plates with separate colonies

Exam Tip: A **log scale** is used because cell numbers increase over several orders of magnitude; during the log phase the line is straight. Explain each phase: lag (cells adjusting, making enzymes), log (plenty of nutrients, cells dividing at their maximum rate), stationary (birth rate equals death rate as nutrients run low and waste builds up), death (more cells die than are produced).

CP13: Isolating Bacteria from a Mixed Culture by Streak Plating



Single colonies in the last streak area

Streak plate

- Flame loop between streaks
- Each streak drags from last
- Cells thinned out each time
- Colony = one cell's offspring

Apparatus

- Mixed broth culture of two non-pathogenic bacteria that form different-looking colonies (e.g. *Micrococcus luteus*, yellow, and *E. coli* K-12 or *Bacillus subtilis*, cream/white)
- Sterile nutrient agar plates, wire inoculating loop, Bunsen burner, disinfectant, adhesive tape, marker pen, incubator at 25 °C

Method

1. Disinfect the bench and work near a Bunsen flame. Label the base of the plate.
2. Flame the loop until red hot and let it cool. Flame the neck of the culture bottle, take a loopful of mixed culture, re-flame the neck and replace the lid.
3. Lift the plate lid slightly and streak the loop back and forth across one edge of the agar (area 1).
4. Flame and cool the loop. Turn the plate about 90° and drag the loop from area 1 into a fresh area (area 2) in a new set of streaks.
5. Repeat for areas 3 (and 4), flaming the loop each time, so that cells are spread more thinly in each area.
6. Tape the lid with two short strips, invert the plate and incubate at 25 °C for 48 hours.
7. Identify separate colonies of each type. To obtain a pure culture, transfer one well-separated colony aseptically onto a fresh plate or into sterile broth.

Safety

- Aseptic technique throughout; do not open plates after incubation unless subculturing under teacher supervision.
- Autoclave all plates and cultures before disposal; wash hands.

Analysis

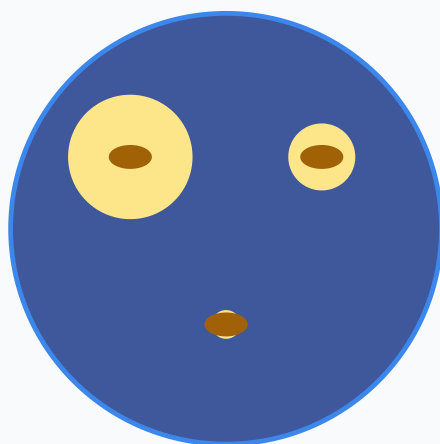
- Describe the colonies of each species (colour, shape, edge, size).
- In the last streaked area, colonies should be well separated; each colony grew from a single cell (or small group), so it contains only one type of bacterium.
- Selective media (containing substances that only some bacteria can tolerate or use) can help isolate a particular species.

Sources of Error

Error	Effect	Improvement
Loop not flamed between areas	Too many cells carried over; no single colonies	Flame and cool the loop before each new area
Hot loop touches bacteria or agar	Cells killed; agar damaged	Cool the loop for a few seconds (or touch the agar edge) before use
Contamination from air or bench	Unexpected colonies	Lift lids only slightly; work near the flame

Exam Tip: Explain why streak plating works: each new streak picks up only a few cells from the previous one, so bacteria are diluted across the plate until single cells are left. Each single cell divides to form a separate colony of genetically identical cells, which can be subcultured to give a **pure culture**.

CP14: Effect of Gibberellin on Amylase Production in Germinating Cereals (Starch Agar Assay)



Starch agar flooded with iodine

Half-grains (no embryo)

- Blue-black = starch remains
- Clear zone = starch digested by amylase
- Larger zone at higher gibberellin concentration

Apparatus

- Barley (or wheat) grains, scalpel, white tile, forceps
- Dilute sodium hypochlorite (bleach) solution for surface sterilising, sterile distilled water
- Gibberellic acid solutions of different concentrations, plus a water-only control
- Sterile starch agar plates, iodine solution, incubator at about 25 °C, ruler

Method

1. Cut each grain in half across its width. Keep the halves **without the embryo** (the embryo is the natural source of gibberellin).
2. Surface-sterilise the half-grains in dilute bleach for a few minutes, then rinse well in sterile water.
3. Soak groups of half-grains in each gibberellin concentration (and in water as a control) for a fixed time.
4. Using sterile forceps, place the half-grains cut face down on labelled starch agar plates, spaced apart. Half-grains with the embryo can be included as a comparison.
5. Tape the lids and incubate at about 25 °C for 24–48 hours.
6. Remove the grains, flood each plate with iodine solution and pour it off. Measure the diameter of the clear zone around each grain.

Safety

- Bleach is corrosive/irritant — wear eye protection and gloves.
- Take care with the scalpel; iodine stains skin.
- Treat plates as possible microbial cultures: autoclave before disposal.

Analysis

- Calculate the area of each clear zone (πr^2) and the mean for each concentration.
- Plot mean area against gibberellin concentration. Larger clear zones show more amylase was produced and secreted.
- Explain: gibberellin causes the **aleurone layer** to switch on transcription of the amylase gene (it leads to the breakdown of DELLA proteins that normally block transcription factors).
Amylase diffuses out and hydrolyses the starch, so iodine gives no blue-black colour there.

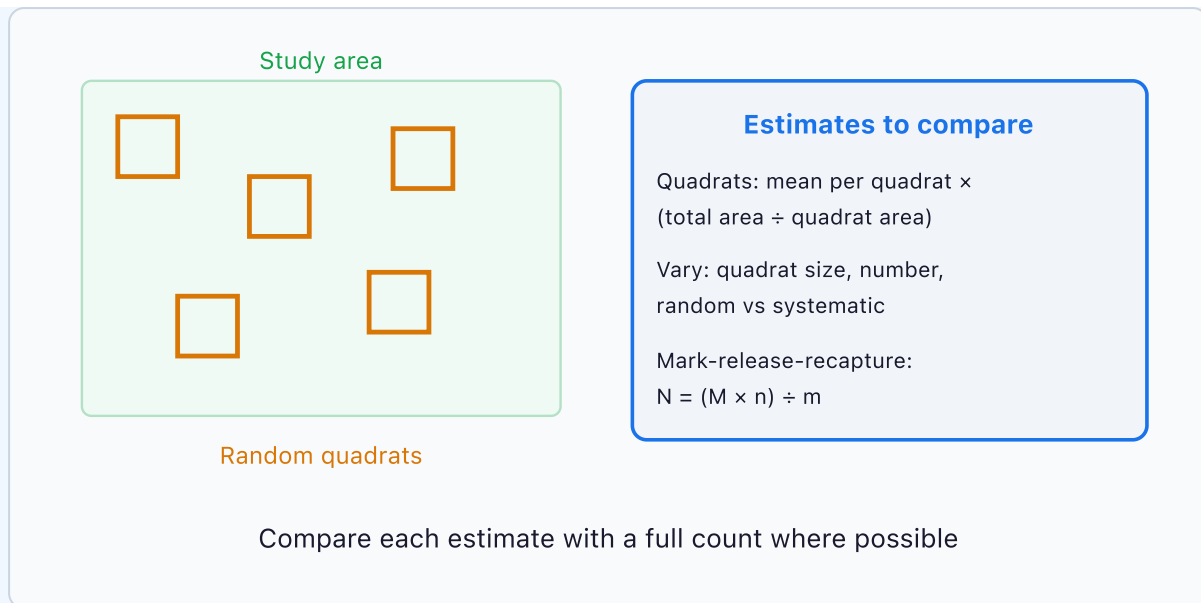
Sources of Error

Error	Effect	Improvement
Microbial contamination	Microbial amylases also digest starch	Surface-sterilise grains; use sterile agar and forceps
Zones not circular	Area inaccurate	Measure two diameters at right angles and average
Grains of different sizes	Different amounts of aleurone tissue	Use grains of similar size; use several per concentration

Exam Tip: Explain the controls: half-grains **without an embryo** are used so the only gibberellin is the gibberellin added; the **water-only** half-grains show how much amylase is produced without it. This practical is evidence that gibberellin regulates gene expression in germinating seeds.



CP15: Effect of Different Sampling Methods on Estimates of Population Size



Apparatus

- Quadrats of different sizes (e.g. 0.25 m × 0.25 m, 0.5 m × 0.5 m, 1 m × 1 m), two tape measures, random number generator
- Pitfall traps or sweep nets and a harmless marker (e.g. non-toxic paint pen) for mobile invertebrates
- Identification keys, recording sheets, calculator

Method

1. Mark out a study area (e.g. 10 m × 10 m of grassland) with two tapes at right angles and choose one easily counted species (e.g. daisies or dandelions).
2. **Random sampling:** use random coordinates to place a quadrat; count the individuals of the species. Repeat for a set number of quadrats (e.g. 10).
3. **Vary the method:** repeat with a different quadrat size, a different number of quadrats, or systematic placement (e.g. every 2 m along lines). Plot a running mean to decide how many quadrats are enough.
4. If the area is small enough, count every individual to find the true population for comparison.
5. **Mobile animals:** estimate a population (e.g. woodlice or ground beetles) by mark-release-recapture: catch and mark a sample (M), release, then catch a second sample (n) and count the marked individuals in it (m).

Safety and Ethics

- Fieldwork risk assessment (weather, terrain, water, stinging plants, ticks). Wash hands after fieldwork.
- Handle animals gently, use marks that do not harm them or make them more visible to predators, check traps regularly and return animals where they were found; avoid trampling the habitat.

Analysis

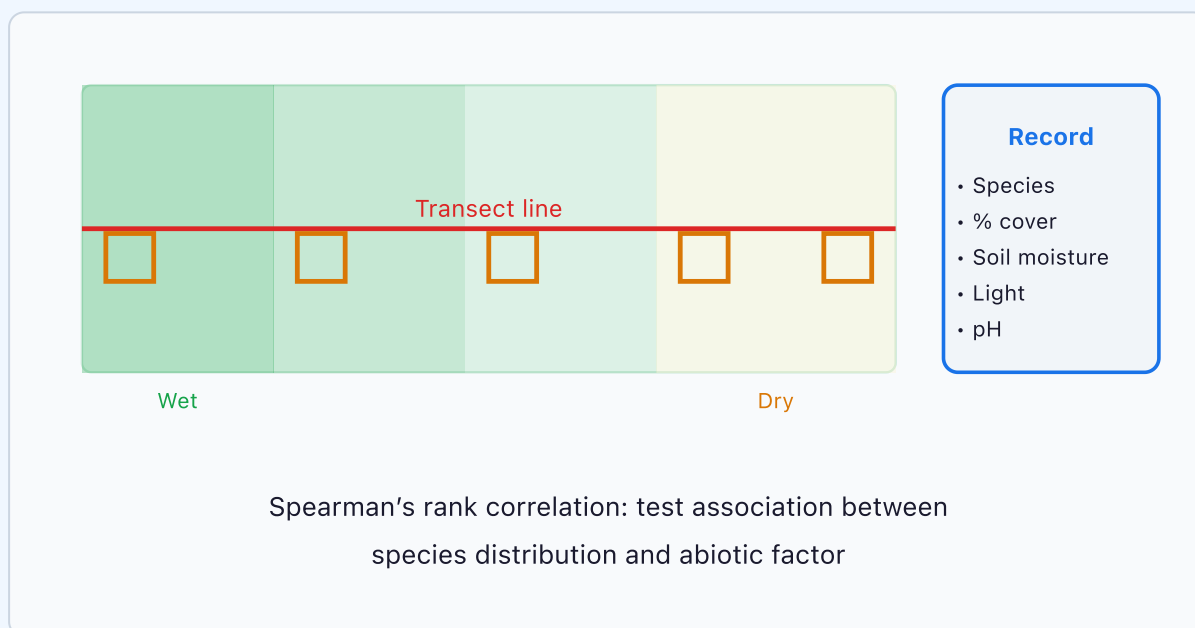
- Population estimate from quadrats = mean number per quadrat $\times$ (area of study site $\div$ area of one quadrat).
- Lincoln index: $N = (M \times n) \div m$.
- Compare the estimates from each method with each other and with the true count. Calculate the standard deviation of counts per quadrat, and discuss which method was most accurate and which most precise.

Sources of Error

Error	Effect	Improvement
Non-random quadrat placement	Biased estimate	Use a random number generator for coordinates
Too few quadrats, or quadrats too small for the species	Estimate varies widely; clumped species missed	Use a running mean to choose sample size; choose a quadrat size suited to the species
Marks lost or marked animals more likely to be eaten	m too low; population overestimated	Use durable, inconspicuous marks; keep the time between samples short
Individuals on the quadrat edge	Inconsistent counts	Agree a rule, e.g. count those on two sides only

Exam Tip: Be ready to justify a sampling method: **random** sampling where conditions are fairly uniform, **systematic** sampling (transects) where conditions change across the area, quadrats for plants and slow-moving animals, and mark-release-recapture for mobile animals. Know the assumptions of mark-release-recapture: marking does not affect survival, marked animals mix fully, and there is no significant birth, death or migration between samples.

CP16: Effect of One Abiotic Factor on the Distribution or Morphology of One Species



Apparatus

- Tape measure (30 m), quadrats (0.5 m × 0.5 m)
- Identification keys, recording sheets
- Abiotic measurement tools: soil moisture meter, light meter, pH probe, thermometer

Method

1. Lay a transect line across a habitat where an environmental gradient exists (e.g. from stream to drier ground, from tree canopy to open).
2. Place quadrats at regular intervals (e.g. every 2 m) along the transect.
3. At each quadrat, record the abundance of the chosen species (percentage cover or number of individuals).
4. Measure abiotic factors (soil moisture, light, pH, temperature) at each point.
5. Record all data in a table.
6. **Morphology alternative:** measure a feature of the one species (e.g. leaf length or leaf area of ivy or plantain) at points where the abiotic factor differs (e.g. light intensity in shade and in the open).

Safety

- Fieldwork risk assessment (terrain, weather, working near water).
- Wash hands after handling soil.

Analysis

- Plot a kite diagram or bar chart showing species distribution along the transect.
- Use Spearman's rank correlation to test for a significant association between an abiotic factor and species distribution.
- If the calculated r_s exceeds the critical value at $p \leq 0.05$, the correlation is significant.

Sources of Error

Error	Effect	Improvement
Too few sampling points	Pattern not detected	Sample at close intervals; many quadrats
Abiotic factors change during day	Measurements not comparable	Take all measurements at same time of day
Misidentification of species	Inaccurate data	Use identification keys; take photos

Trampling affects habitat	Alters results in subsequent quadrats	Walk carefully; avoid stepping on quadrat area
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Exam Tip: Spearman's rank is used for correlation between two sets of data where one or both are ordinal or not normally distributed. If r_s is close to +1 or -1, there is a strong positive or negative correlation. Always compare to the critical value at the chosen significance level ($p \leq 0.05$).